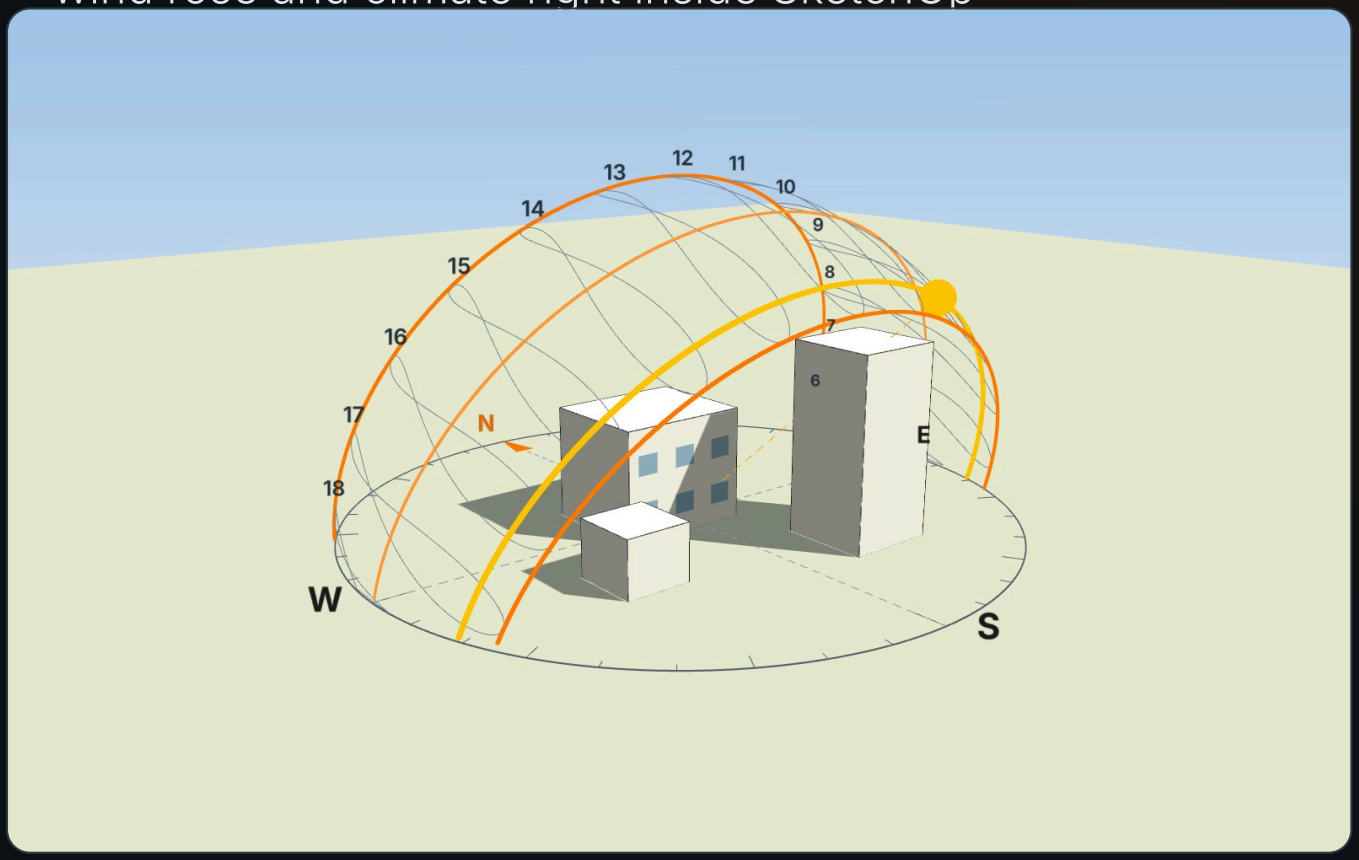
SKETCHUP PLUGIN · USER GUIDE

BETAGON SUN

Sun path, shadows, sun hours,
wind rose and climate right inside SketchUp



Copyright **Betagon** · Author: **Nguyễn Văn Thanh**

betagonplugin.com

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1 Introducing Betagon Sun

Betagon Sun is a **free** SketchUp plugin that helps architects, students and interior designers understand the sun, sunlight and wind on site from the very first sketch: where the sun travels, which sides get the most sun, how deep an overhang must be, where the prevailing wind comes from and what each month's climate is like. Every result can be exported to a one-page PNG report for presentations and design reports.

- Pick from 70 Vietnamese cities and 23 world cities, or type coordinates. **By IP** finds this computer's location.
- Apparent sun path diagram with hour lines, solstices and equinoxes.
- Drag the time bar or press ▶ to play the SketchUp shadows through the day.
- Clear-sky irradiance on 8 facade orientations and the roof, plus an overhang and louvre calculator.
- Draw a 3D sun path dome and a 3D wind rose right in the model.
- Sun hours analysis on floors, walls, roofs and yards, with shadows of the surrounding buildings.
- Wind roses for 5 Vietnamese climate regions, or 3 years of real wind data at your coordinates.
- Export an A4 report or single charts as PNG. Light / dark theme, English / Tiếng Việt.

> **Requirements**
> SketchUp 2021 to 2027 on Windows or macOS. The plugin is free and needs no licence. **By IP**, **Get real data** and the **Climate** tab need Internet; everything else works fully offline.

2 Install, update and uninstall

Install through Betagon Center (recommended)

1. Open the **Betagon Center** app in SketchUp (download it from **betagonplugin.com** if you don't have it).
2. Find **Betagon Sun** in the plugin list and click **Install**. It is free, so it installs right away with no purchase or trial.
3. The **Betagon Sun** toolbar appears. If not, open **View › Toolbars** and tick **Betagon Sun**.

Install from the .rbz file

In SketchUp open **Extensions › Extension Manager**, click **Install Extension**, pick **Betagon_Sun_V1.0.0.rbz** and confirm.

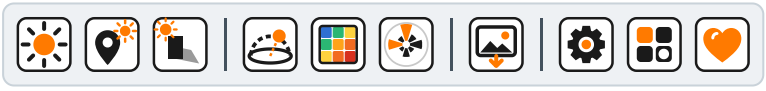
Updating

Betagon Center tells you when a new version is out. Installing it over the running version **needs no SketchUp restart**: the plugin reloads itself, the open panel switches to the new version and keeps your location, date and time.

Uninstall

Open **Extensions › Extension Manager**, find **Betagon Sun** and click **Uninstall**. Objects the plugin drew in a model (sun path, wind rose, analysis results) stay in that .skp file until you delete them.

3 Toolbar and menu



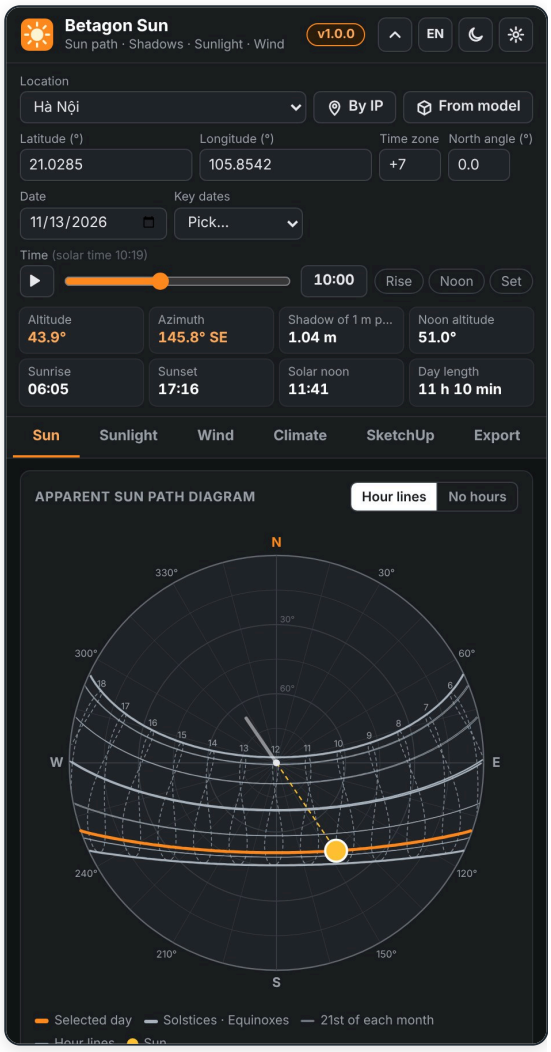
	Button	What it does
	Betagon Sun Panel	Opens the main panel: location, date and time, charts, sunlight, wind, climate, shadows, export. If the panel is open, brings it to the front.
	Locate by IP	Gets coordinates, city and time zone from this computer's IP (needs Internet) and writes them into the model shadows.
	Shadows On / Off	Turns SketchUp shadows on or off with the location, date and time set in Betagon Sun.
	Draw 3D Sun Path	Draws the sun path dome, compass and sun ball around the selection (or the model centre).
	Sun Hours Analysis	Select faces (floors, walls, roofs, yards...) and click: coloured by hours of direct sun.
	Draw 3D Wind Rose	Draws the whole-year wind rose of the location on the ground around the selection.
	Export Results as Image	Opens the panel on the Export tab to save the report, a chart or the shadowed model view as PNG.

	Settings – About	Language, theme, tutorial videos, community, contact, donate, copyright.
	Betagon Center	Opens Betagon Center to manage and update plugins. No Center yet: click to download and install it.
	Support the Author	Shows the donation QR code. Thank you for supporting Betagon!

Menu: **Extensions › Betagon › Betagon Sun** holds the panel, Locate by IP, Shadows On / Off and the draw, analyse and export commands. Hover a button to see a short description.

4 Quick start in 5 steps

1. Click **Betagon Sun Panel**. The first time it opens on **Hà Nội**, today, 10:00.
2. Pick a location from the list, click **By IP** for this computer's location, or **From model** if the model already has a Geo-location.
3. Enter the model's **North angle** (the angle between true north and SketchUp's green axis). If the model is drawn with north along the green axis, leave 0.
4. Pick a date, then drag the time bar or press ▶: charts, figures and SketchUp shadows follow. **Rise Noon Set** jump straight to those times.
5. Browse the Sun, Sunlight, Wind and Climate tabs; draw the 3D sun path and analyse sun hours on the SketchUp tab; finally save a PNG report on the **Export** tab.



The Betagon Sun panel, default dark theme

Shadows follow the panel

While **Sync shadows** is on (default), every change of location, date, time or north angle in the panel is written straight into the model's **Window › Shadows**. No need to edit Shadows by hand.

One Undo step

Every sun path, wind rose or sun hours analysis is one Ctrl + Z step. Moving the sun ball by dragging the time bar is merged into one step, so it never floods the Undo history.

5 The Betagon Sun panel: location, date, time and figures



The top of the panel is always visible and shared by every tab

Title bar

- The **up arrow** button collapses the location and figures to give the charts more room; click again to expand.
- **VI / EN** switches language, the **moon** button switches light / dark theme, the **gear** button opens Settings – About.

Location

- **List**: 70 Vietnamese cities and provinces and 23 major world cities.
- **By IP**: coordinates, city and time zone from your Internet provider. IP locations are approximate, so double-check.
- **From model**: reads the location set in the model (Geo-location or Shadows).

Field	Meaning
Latitude, Longitude	Decimal degrees. Positive = northern hemisphere / east. Hà Nội ≈ 21.03 / 105.85. Typing coordinates switches the location to *Custom*.
Time zone	Offset from UTC. Vietnam = +7.
North angle	Angle (degrees, clockwise) from SketchUp's green axis to true north. Keeps shadows, the 3D sun path and wind rose aligned with the building.
Date, Key dates	Any date, or quickly: Now, Equinox Mar 20, Solstice Jun 21, Equinox Sep 23, Solstice Dec 21.
Time	Drag the time bar (5-minute steps) or type a time (e.g. *14:30*, *7h30*). The grey label shows the true solar time at the location.
▶ Rise · Noon · Set	▶ plays time continuously so you can watch shadows move; Rise / Noon / Set jump to sunrise, solar noon and sunset.

8 live figures

Altitude: sun angle above the ground (0° = horizon, 90° = overhead).

Azimuth: sun direction, clockwise from north.

Shadow of 1 m pole: shadow length of a 1 m tall pole right now.

Noon altitude: the highest sun of the selected day.

Sunrise / Sunset: local time in the time zone entered.

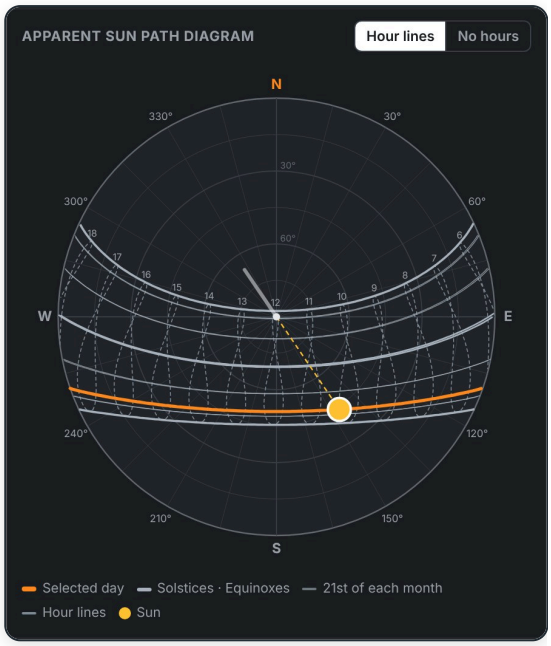
Solar noon · Day length: when the sun is highest and total daylight.

6 Sun tab: the apparent sun path diagram

The diagram looks at the sky from above: the **centre** is straight overhead, the **outer ring** is the horizon and the inner rings are 30° and 60° altitude. **N E S W** mark the four directions, the degrees around the ring are azimuths.

- **Bold orange line**: path of the selected day, the **yellow dot** is the sun now, joined to the centre by a dashed line.
- **Grey lines**: paths on the solstices, equinoxes and the 21st of each month, showing how high or low the sun goes by season.
- **Hour lines** (figure-of-eight analemmas): the sun's position at the same clock time over the year. **No hours** hides them for a cleaner view.

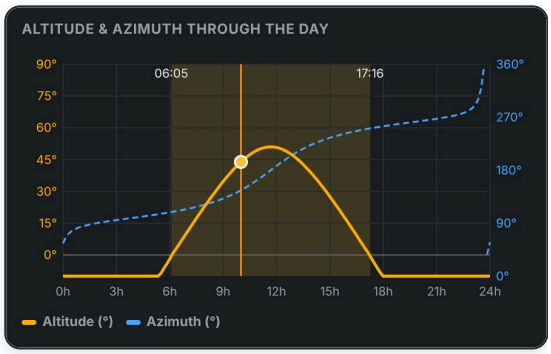
> **Reading it**
> Where is the sun at 15:00 on June 21? Find where hour line 15 crosses the June solstice path. Its direction is where sunlight comes from; its distance from the centre tells whether the sun is high or low.



Apparent sun path, Hà Nội, Nov 13 at 10:00

Altitude and azimuth through the day

The lower chart plots **altitude** (orange, left axis) and **azimuth** (dashed blue, right axis) against time. The light band is daylight and the vertical line is the selected time.



Altitude and azimuth through the day

7 Sunlight tab: irradiance, sunrise and sunset, overhangs

Clear-sky irradiance by orientation

The chart shows how much solar energy (W/m²) each side of a building receives hour by hour on the selected day, using the ASHRAE clear-sky model. The table below sums it up for the roof and 8 facade orientations:

- **Peak (W/m²)** and **At**: when the sun is strongest.
- **Sun hours on it**: how long that side is in the sun.
- **Daily (kWh/m²)**: the bigger it is, the hotter that side, so shade or insulate it first.

The orientation with the most sun is highlighted in orange. In Vietnam the **west** side is usually the hottest on summer afternoons, while the **south** side gets the most sun in winter.



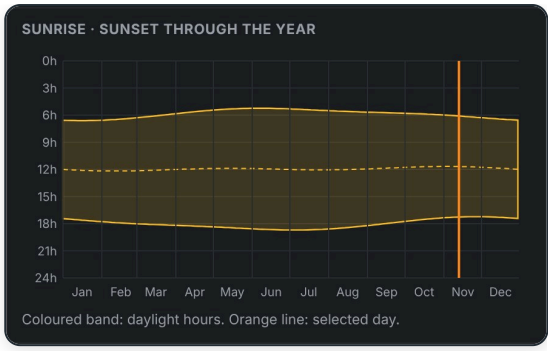
Side	Peak (W/m²)	At	Sun hours on it	Daily (kWh/m²)
Roof (flat)	847	11:40	11 h 10 min	5.69
North	116	11:40	0 h 00 min	0.85
East	722	08:10	5 h 40 min	3.13
South	**750**	**11:40**	**11 h 10 min**	**5.78**
West	721	15:10	5 h 30 min	3.13

ASHRAE clear-sky model, selected day. The bigger the daily total, the hotter that side: shade or insulate it.

Irradiance by orientation with summary table

Sunrise · sunset through the year

The band shows daylight hours for every day of the year; the orange line is the selected day. Use it to compare long and short days and pick the time for renders and photos.



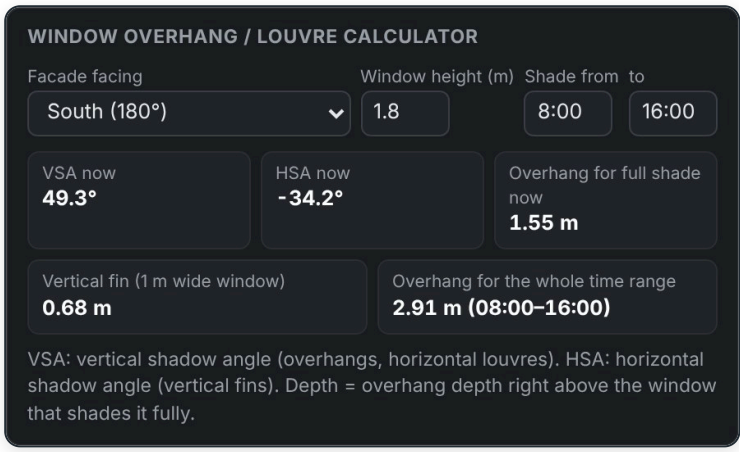
Sunrise · sunset through the year

Window overhang / louvre calculator

Pick the **facade** with the window, enter the **window height** and the **time range** to shade (default 8:00 – 16:00). The plugin calculates at once:

- **VSA** (vertical shadow angle): for overhangs and horizontal louvres. **HSA** (horizontal shadow angle): for vertical fins.
- **Overhang for full shade now**: depth of an overhang right above the window that shades it fully at the selected time.
- **Overhang for the whole time range**: depth needed to shade it fully for the whole range.
- **Vertical fin**: fin depth for a 1 m wide window.

If the sun never reaches that facade the plugin says *No sun on this facade*.



Example: 1.8 m high window facing south, shaded 8:00 – 16:00

Example: south window, Hà Nội, Nov 13 at 10:00

Window height 1.8 m. The plugin gives **VSA = 49.3°**. Overhang depth right above the window for full shade = window height ÷ tan(VSA) = 1.8 ÷ tan 49.3° ≈ **1.55 m**, matching *Overhang for full shade now*.

Full shade from 8:00 to 16:00 needs **2.91 m**, which is a lot. In practice split it up: a 0.8 – 1.0 m overhang with horizontal louvres, or raise the overhang above the window head and add side fins (based on **HSA**). Switch the date to the June 21 and December 21 solstices to check both seasons before fixing the sizes.

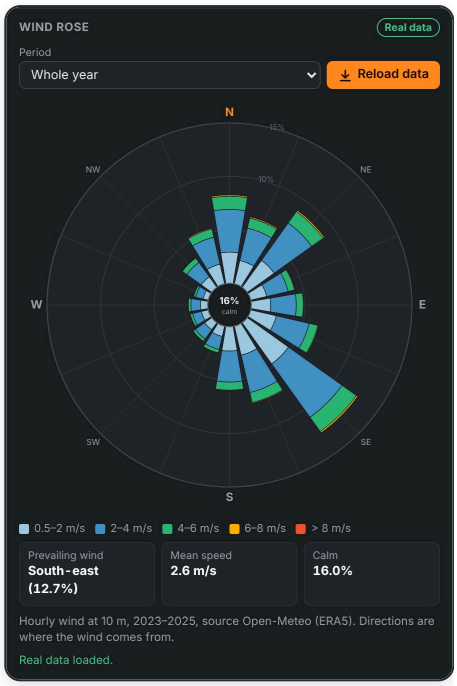
Tip: a large VSA (high sun) means a short overhang is enough; a small VSA (low, slanting sun, usually on east and west sides) needs a very deep overhang, so use vertical fins, blinds or trees instead.

8 Wind tab: estimated and real wind roses

The wind rose shows **where the wind comes from**, how often and how strong at the location. Each petal is one of 16 directions; the longer the petal, the more often the wind blows from there (% rings). Colours inside a petal are wind speeds:

0.5–2 m/s | 2–4 | 4–6 | 6–8 | > 8 m/s

The number in the centre is the share of **calm**. Below the chart you see the **prevailing wind**, **mean speed** and the number of data hours.



Wind rose from real data, whole year

Two data sources

- **Regional estimate** (default, offline): monsoon patterns of 5 regions: Northern, North-Central, South-Central coast, Central Highlands and Southern Vietnam. For a quick reference only.
- **Real data**: click **Get real data** to download hourly wind of the last 3 years at your coordinates (Open-Meteo / ERA5, free, needs Internet). The data is cached on your computer, so it opens instantly next time.

Period filters the rose by whole year, dry season (north-east monsoon), wet season (south-west / south-east monsoon) or single months.

> **Use it for**
> Placing openings towards the prevailing summer wind for natural ventilation; screening the cold winter wind (north-east in northern Vietnam).

Regional estimates

Without real data the plugin picks the region from the location's latitude and longitude and builds the rose from that region's monsoon pattern (wet-season months lean towards the summer monsoon, dry-season months towards the winter monsoon):

Region	Range	Winter monsoon (dry season)	Summer monsoon (wet season)
Northern Vietnam	Latitude 19.5° and above	North-east, North	South-east, South-south-east
North-Central Vietnam	Latitude 15° – 19.5°	North, North-east, North-west	East, East-south-east; hot dry south-west wind
South-Central coast	Latitude 11° – 15°, coastal	North-east, North-north-east	South-west, West-south-west, East-south-east
Central Highlands	Latitude 11.6° – 15°, longitude below 108.6°	North-east, East-north-east	West-south-west, South-west, West
Southern Vietnam	Latitude below 11°	East, East-south-east	South-west, West-south-west

Outside Vietnam there is no built-in estimate: click **Get real data**. Estimates are a quick reference only; for ventilation design use real data at your coordinates.

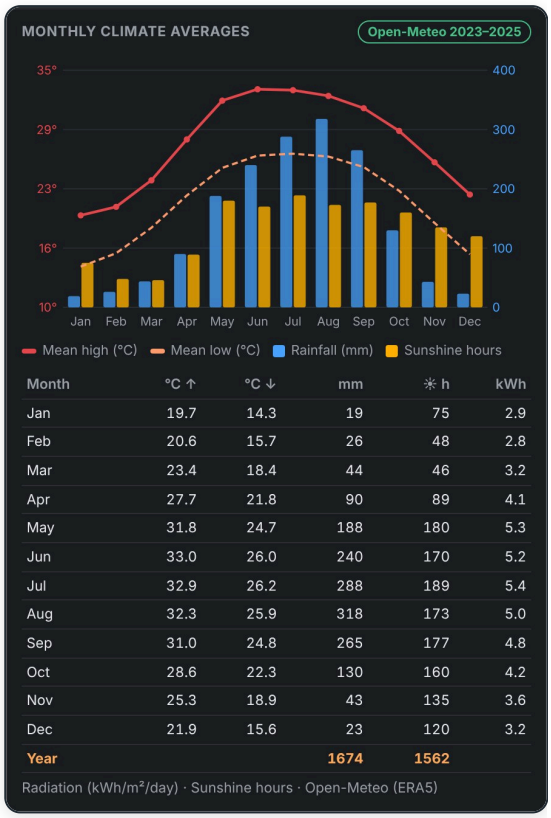
9 Climate tab: temperature, rain and sunshine by month

The Climate tab sums up the average climate of the last 3 years at the selected coordinates. The first time it says *No climate data for this location yet*: click **Get real data** right in the tab (or on the Wind tab, it is the same download) to fetch it from Open-Meteo.

- **Chart**: blue bars are rainfall (mm), yellow bars are sunshine hours, the red line is the mean high and the dashed orange line the mean low temperature.
- **Table**: each month with mean high and low (°C), rainfall (mm), sunshine (h) and radiation (kWh/m²/day); the **Year** row holds total rain and sunshine.

Use it to choose materials, shading, roof drainage and planting, or in your design statement.

> **Note**
> The figures come from the ERA5 reanalysis via Open-Meteo, good enough for early design. For official submissions use your national standard or the nearest weather station.



Month	°C ↑	°C ↓	mm	☼ h	kWh
Jan	19.7	14.3	19	75	2.9
Feb	20.6	15.7	26	48	2.8
Mar	23.4	18.4	44	46	3.2
Apr	27.7	21.8	90	89	4.1
May	31.8	24.7	188	180	5.3
Jun	33.0	26.0	240	170	5.2
Jul	32.9	26.2	288	189	5.4
Aug	32.3	25.9	318	173	5.0
Sep	31.0	24.8	265	177	4.8
Oct	28.6	22.3	130	160	4.2
Nov	25.3	18.9	43	135	3.6
Dec	21.9	15.6	23	120	3.2
Year			1674	1562	

Radiation (kWh/m²/day) · Sunshine hours · Open-Meteo (ERA5)

Monthly climate averages, Hà Nội

Where is the data stored?

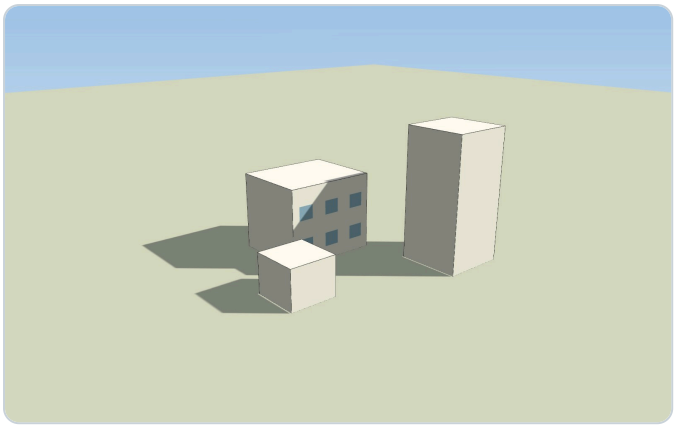
Each downloaded location is cached in the Betagon Sun settings folder on your computer (*%APPDATA%/Betagon/Sun/climate* on Windows), so reopening needs no download. Click **Reload data** to refresh it.

10 SketchUp tab: shadows, 3D sun path, sun hours, 3D wind rose

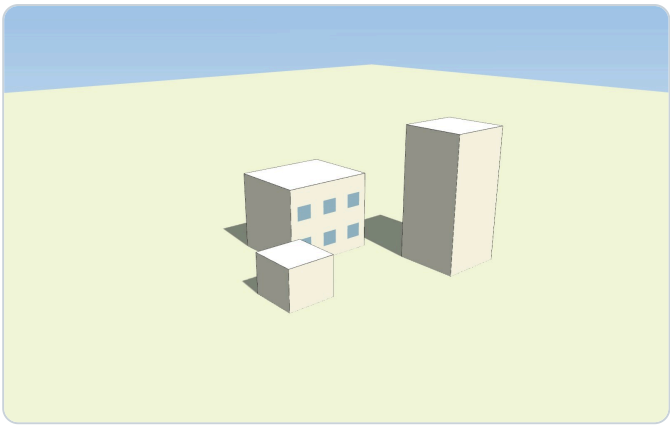
SketchUp shadows

- **Sync shadows**: writes location, date, time and north angle into the model Shadows whenever you change them in the panel.
- **Show shadows**: shadows on / off in the view (same as the toolbar button).
- **Playback speed** 1× / 2× / 4×: the speed of the ▶ button.





Shadows at 9:00, Nov 13, Hà Nội



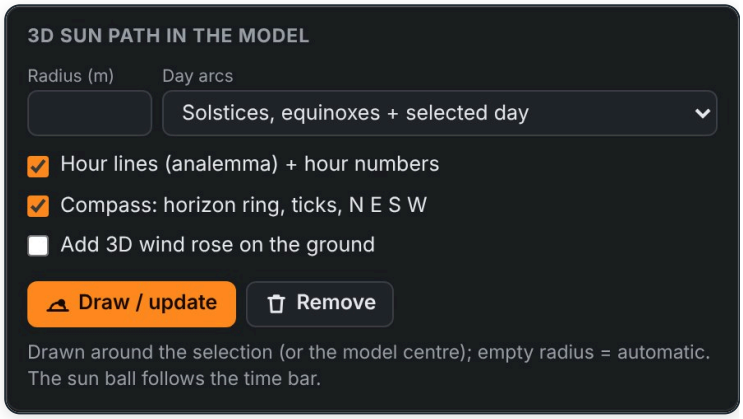
Drag the time bar to 12:00: shadows follow

3D sun path in the model

Select the building (or nothing for the model centre) and click **Draw / update** or the button. The plugin draws one group with:

- A compass: horizon ring, ticks every 10°, letters **N E S W** and an orange north arrow.
- Day arcs of your choice: *solstices, equinoxes + selected day*, *21st of 12 months* or *selected day only*.
- Hour lines (analemmas) with hour numbers; the sun ball moves as you drag the time bar.
- Optional **3D wind rose on the ground**.

Leave **Radius** empty to size it from the building automatically. Click again to update after changing the location or date; **Remove** deletes it. Everything sits on its own *Betagon Sun* tag so you can hide it quickly.



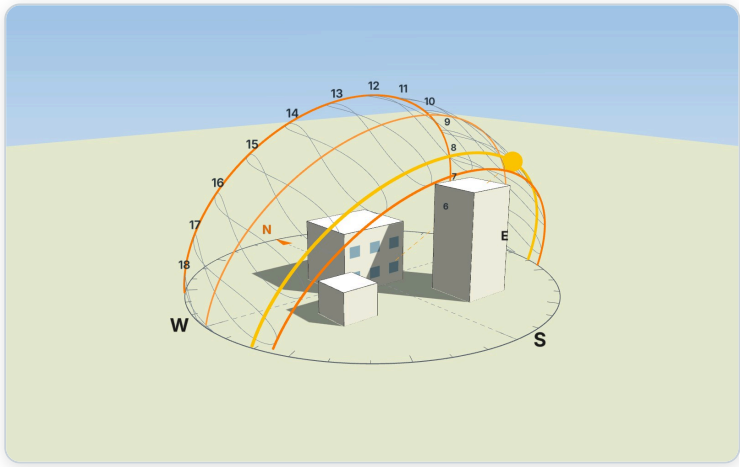


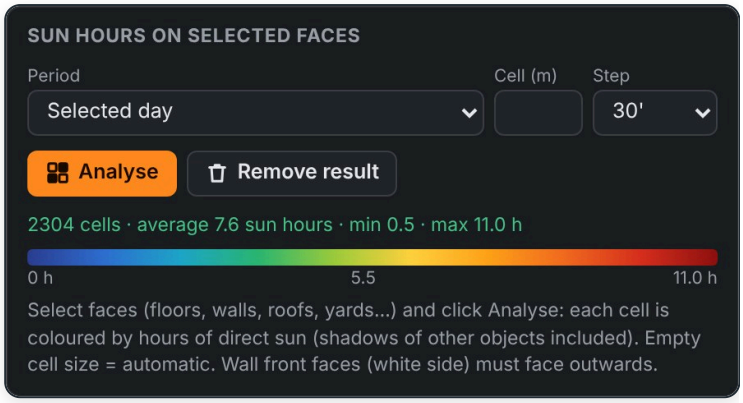
Illustration: 3D sun path dome, the sun ball at 10:00 on Nov 13 and the matching shadows

What is the 3D sun path for?

- Check which rooms get morning sun and which windows take direct afternoon sun: look out from the room towards the sun ball.
- Place solar panels, skylights or drying yards: drag the time bar through the June and December solstices and use the spots that stay sunny.
- Present your design: the dome and compass sit on the *Betagon Sun* tag, so show them for concept diagrams and hide them for renders.

Sun hours on selected faces

1. Select the faces to analyse: floors, yards, walls, roofs... (Groups / Components with faces work too).
2. Choose the **Period**: selected day, year average (21st of each month), summer (May 21 – Aug 21) or winter (Nov 21 – Feb 21).
3. **Cell** (m): empty = automatic; smaller is finer but slower. **Step** 15' / 30' / 60' is the time between two samples.
4. Click **Analyse**. A progress bar runs; click **Stop** to cancel. SketchUp stays usable while it works.



The result appears right under the buttons

Each cell is coloured by hours of direct sun, **including shadows of other objects** in the model. The 10-step scale goes from dark blue (little sun) to dark red (most sun):



0 h — middle — most

The result is a separate group that you can hide or delete with **Remove result**. The panel reports the number of cells and the average, minimum and maximum sun hours.

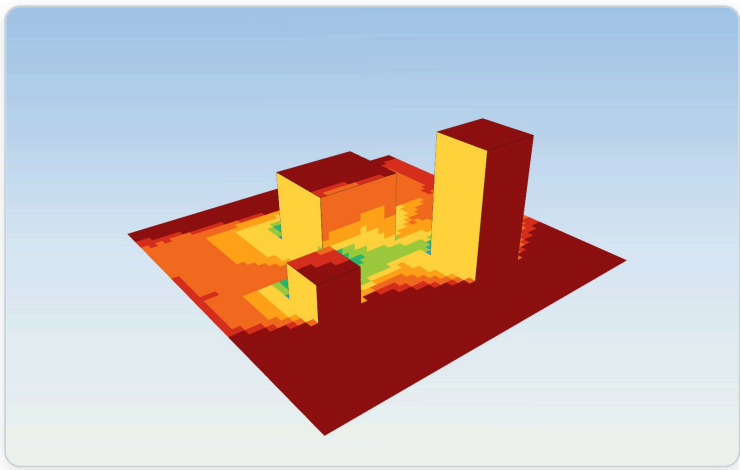


Illustration: sun hours on Nov 13 on the yard, walls and roofs

> **Front faces must face outwards**
> The plugin measures sun on the **front face** (white by default) of each face. Reversed walls (blue-grey back face outside) give wrong results: right-click › **Reverse Faces** before analysing.

3D wind rose on the ground

Click **Draw 3D Wind Rose** (or tick *Add 3D wind rose* when drawing the sun path) to draw the whole-year wind rose on the ground around the selection. If real data was downloaded the rose uses it; otherwise the regional estimate.

The rose follows the model's **North angle**, so on a site plan you see at once which rooms face the wind. Petal colours match the Wind tab.

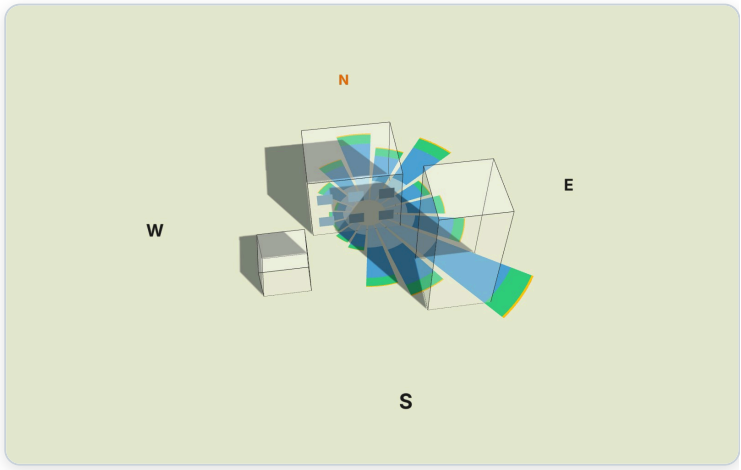


Illustration: 3D wind rose (Hà Nội, whole year) on the ground around the buildings

Remove all Betagon Sun objects

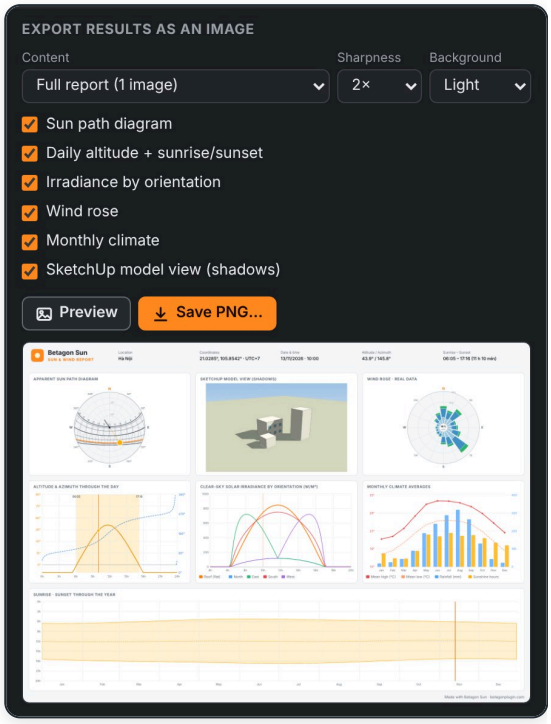
The last button on the SketchUp tab deletes the 3D sun path, 3D wind rose and sun hours

results in one go. Your own model is not touched.

11 Export tab: A4 report and single charts

1. Choose the **Content**: *Full report (1 image)* or a single part: sun path diagram, daily altitude + sunrise/sunset, irradiance by orientation, wind rose, monthly climate, SketchUp model view.
2. For the report, tick the parts to include. **Sharpness** 1× / 2× / 3× (2× is enough for A4 print), **Background** light or dark.
3. Click **Preview** to see the image in the panel.
4. Click **Save PNG…** and choose where to save. The panel shows the file path when done.

SketchUp model view captures the current camera with shadows, so orbit to a nice view before exporting. The toolbar button opens this tab directly.



Export tab with preview



A4 landscape sun & wind report exported by Betagon Sun (light background, 1× sharpness)

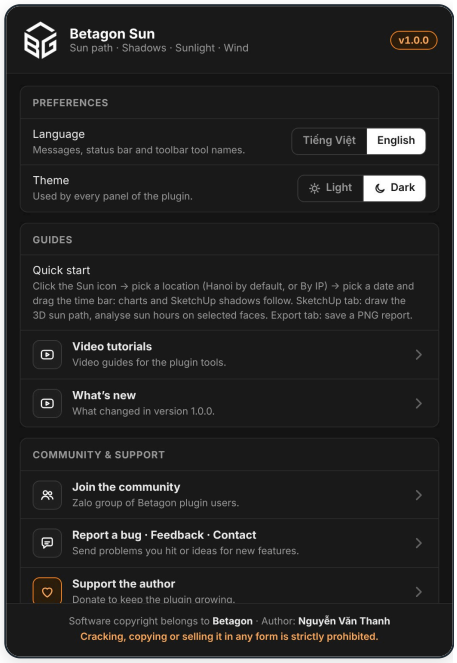
The report starts with location, coordinates, date and time, altitude / azimuth and sunrise – sunset, followed by 6 panels: apparent sun path, model view, wind rose, daily altitude, irradiance by orientation, monthly climate, and the sunrise – sunset band for the whole year. Unticked parts are left out and the remaining panels stretch to fill the page.

12 Settings – About window

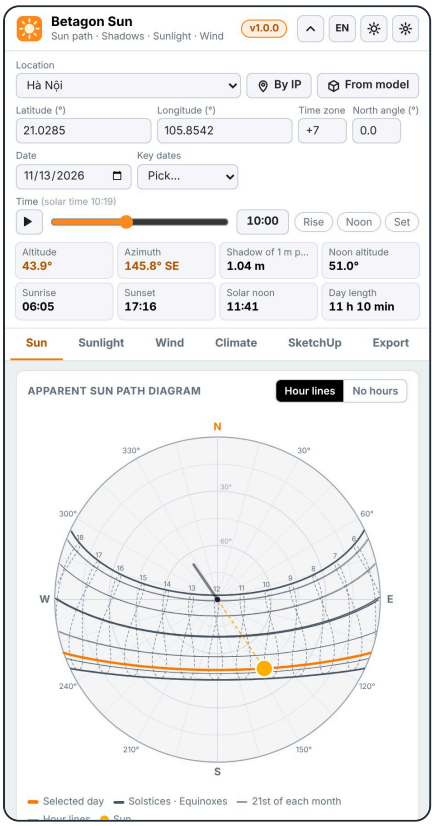
Open it with the ⚙ toolbar button or the gear button in the panel title bar.

- **Language**: English / Tiếng Việt for messages, panels and toolbar names.
- **Theme**: Light / Dark for every panel of the plugin.
- **Guide**: quick how-to, tutorial video, release video.
- **Community & support**: Betagon Zalo user group, bug reports and feedback, support the author, Betagon Center, website.

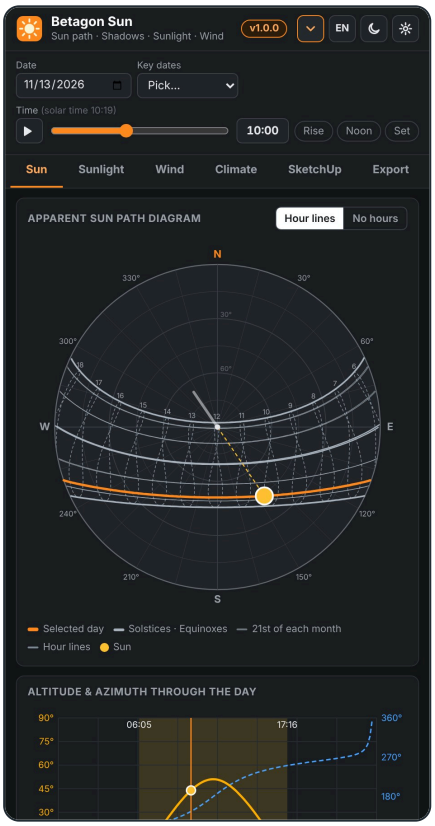
Language and theme are remembered for next time.



Settings – About window



Light theme



Location collapsed

13 Tips, FAQ and troubleshooting

Problem	What to do
By IP gives the wrong city	IP locations follow your Internet provider and can be off (often the provincial capital or a big city). Pick from the list or type exact coordinates (from Google Maps: right-click › the coordinates).

Cannot locate / cannot download data	Check the Internet connection and company firewall. Offline parts (charts, shadows, 3D sun path, sun hours, estimated wind rose) keep working.
Shadows point the wrong way	Check the **North angle**. If the model has a Geo-location, click **From model**. Turn on **Sync shadows** on the SketchUp tab.
Changing the location does not change the shadows	A model with a **Geo-location** (Add Location) keeps its real location; the plugin only writes the date and time. Click **From model** so the panel uses that location, or clear the Geo-location to try other places.
SketchUp Shadows time differs from the panel	The plugin writes local time in the time zone entered; SketchUp shows the model's time zone. Set the right time zone (+7 for Vietnam) and they match.
Sun hours analysis is slow	Use a larger **Cell** (e.g. 1 m instead of 0.5 m), a **Step** of 30' or 60', or fewer faces. You can click **Stop** at any time.
"Select at least one face"	Select faces, or Groups / Components containing faces, and click Analyse again.
Walls come out all blue	The walls are reversed: right-click › **Reverse Faces** so the front faces face outwards.
Remove everything the plugin drew	SketchUp tab › **Remove all Betagon Sun objects from the model**, or hide the *Betagon Sun* tag.
Model view in the report is dark / wrong angle	Orbit to the view you want, turn shadows on, then click Preview again before saving.
New version installed but nothing changed	Close and reopen the Betagon Sun panel. If that does not help, restart SketchUp.

Design tips

Use the **June 21 solstice** to check harsh west sun and summer overhang depths; the **December 21 solstice** to check overshadowing between buildings (low winter sun, longest shadows); and the **year average** when analysing sun hours on gardens and playgrounds.

14 Support and contact

Website: betagonplugin.com – download Betagon Center and Betagon plugins.

Zalo community: zalo.me/g/rkagf7cvowt17pmlo0f9 – questions and tips with other users.

Fanpage: facebook.com/Betagon.SketchUp – tutorial videos and news.

Bug reports – feedback: open Settings – About › *Report a bug · Feedback · Contact*. A screenshot and your SketchUp version help a lot.

Support the author: Betagon Sun is free. If it helps you, you can donate with the ❤ button to keep new tools coming.

Copyright

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